Multi-Objective Optimization for Fairness and Complexity in Machine Learning

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Abstract

Our report describes a methodical approach towards implementing MOO for prediction models using COMPAS dataset. Through the use of NSGA-II algorithm (from the Python library Pymoo) along with the NSGAIISampler in Optuna, we simultaneously maximize prediction accuracy, minimize demographic parity, and the complexity of our model. This method allows for revealing tradeoffs between these three goals and ultimately finding an optimal compromise.

1 Introduction

Modern machine learning models are often optimized solely for predictive performance (e.g., accuracy, AUC). However, in high-stakes domains such as criminal justice, evaluating models across a single dimension is insufficient. Models must also exhibit fairness across sensitive demographic groups and remain interpretable and computationally lightweight. This project explores the use of evolutionary algorithms to discover optimal trade-offs between these competing priorities, mapping out a multi-dimensional Pareto frontier.

2 Problem Statement and Objectives

The primary problem addressed in this study is the simultaneous optimization of a machine learning pipeline across three inherently conflicting objectives. In standard optimization, improving fairness often degrades accuracy, while increasing accuracy often requires highly complex (and less interpretable) models.

Our multi-objective formulation seeks to minimize the following three objective functions:

1. **Minimize** $(1 - \textbf{Balanced Accuracy})$**:** Balanced accuracy is used instead of standard accuracy to account for class imbalances in the COMPAS dataset. Maximizing balanced accuracy corresponds to minimizing its complement.

2. **Minimize Demographic Parity (DP) Violation:** Fairness is quantified as the absolute difference in the positive prediction rate (recidivism prediction) between the unprivileged group (African-American) and the privileged group (Caucasian).

$$f_2 = |P(\hat{Y} = 1 \mid A = \text{African-American}) - P(\hat{Y} = 1 \mid A = \text{Caucasian})|$$

3. **Minimize Model Complexity:** We define a normalized metric $\in [0, 1]$ representing the computational and structural complexity of the chosen model. Logistic Regression is penalized linearly by its regularization strength, while tree-based models (Random Forest, XGBoost) are penalized based on their number of estimators and maximum tree depth. This objective acts as a proxy for interpretability and deployment cost.

These three objectives are intrinsically conflicting. Achieving zero DP violation often requires severely degrading predictive accuracy, and reaching peak accuracy typically demands high model complexity.

3 Methodologies

To navigate this complex search space consisting of mixed-variable hyperparameters (model type, regularization, learning rate, tree depth, etc.), we employed two distinct NSGA-II based methodologies.

3.1 Method 1: NSGA-II via Pymoo (`moo_nsga2_pymoo.py`)

The first methodology utilizes `pymoo`, a specialized library for multi-objective optimization.

- **Core Components:**
  - `load_and_preprocess_compas()`: Handles the loading and standardized ProPublica filtering of the dataset.
  - `compute_fairness_metrics()`: Calculates the Demographic Parity violation across the sensitive groups.
  - `compute_model_complexity()`: Evaluates the normalized structural complexity of the generated models.
  - `FairnessMOOProblem`: A custom class inheriting from `pymoo.core.problem.Problem` that formally defines the 7 decision variables (including continuous variables like learning rate and discrete variables like model type) and the 3 objectives.
- **Execution:** The algorithm was configured with a population size of 150 and run for 150 generations (evaluating roughly 22,500 candidate models) using Simulated Binary Crossover (SBX) and Polynomial Mutation (PM).

3.2 Method 2: Optuna NSGA-II Sampler (`moo_optuna_comparison.py`)

The second methodology relies on Optuna, a highly efficient hyperparameter optimization framework utilizing its native `NSGAIISampler`.

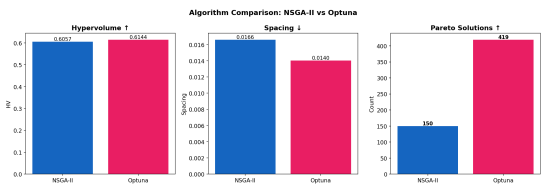
- **Core Components:**
  - While reusing the data loading and complexity metric functions, this script defines a unified `optuna_objective(trial)` function.
  - The objective function utilizes Optuna's `trial.suggest_categorical`, `trial.suggest_float`, and `trial.suggest_int` APIs to dynamically sample the search space.
  - `evaluate_optuna_pareto()`: Extracts the best trials forming the Pareto front and performs out-of-sample evaluation on the test set.
- **Execution:** The study was executed for 2,500 multi-objective trials. While Optuna is generally more sample-efficient through its Bayesian-guided approach, Pymoo provides more rigorous population-based evolutionary dynamics.

4 Results and Visualizations

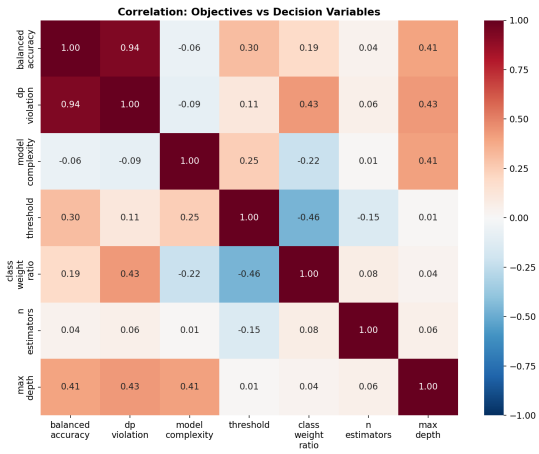
The optimization successfully identified dense Pareto frontiers for both methodologies. The results were parsed to visualize the exact dynamics of the trade-offs.

4.1 Algorithm Comparison and Trade-offs

Figure 1 illustrates the global performance and interactions of the objectives.



(a) Algorithm Comparison (Hypervolume & Spacing)



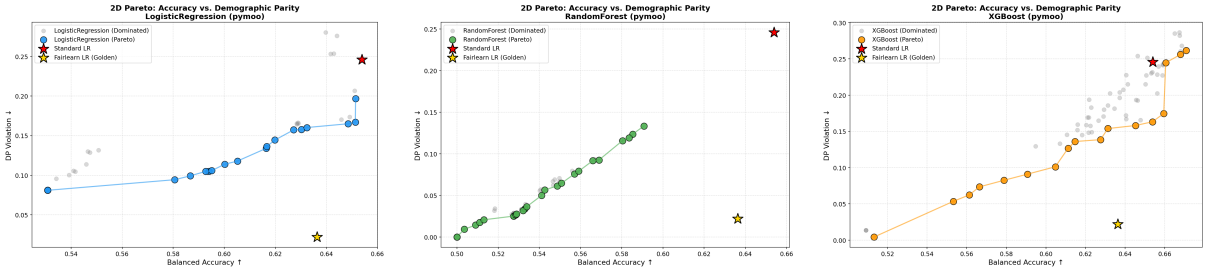
(b) Objective Trade-off Heatmap

Figure 1: Aggregate summaries detailing algorithm performance and inter-objective correlations.

4.2 Pareto Projections

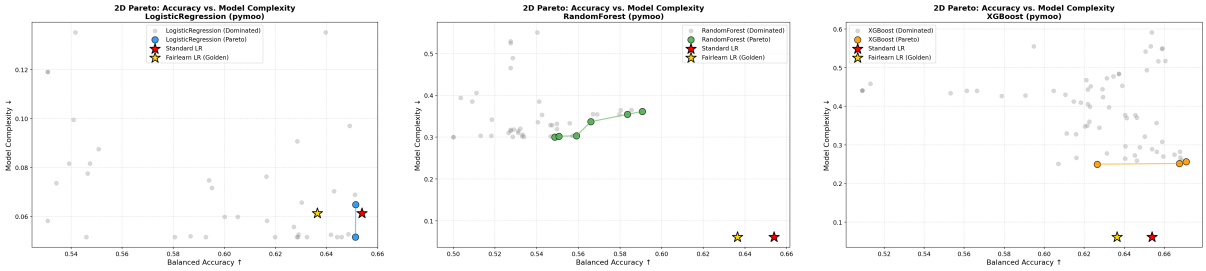
To understand how model complexity affects fairness and accuracy, we project the 3D Pareto frontier into 2D spaces. In these visualizations, sub-optimal (dominated) solu-

tions are plotted as a dimmed background cloud, while the optimal frontier is boldly highlighted.



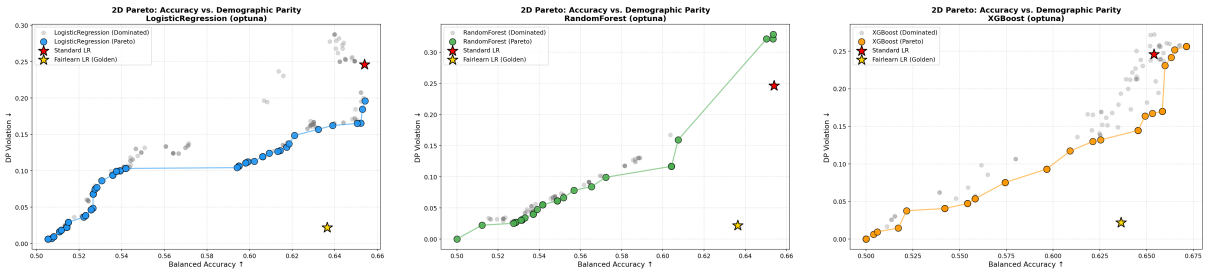
(a) Logistic Regression (b) Random Forest (c) XGBoost

Figure 2: Pymoo NSGA-II: Balanced Accuracy vs. Demographic Parity Violation across three model types.



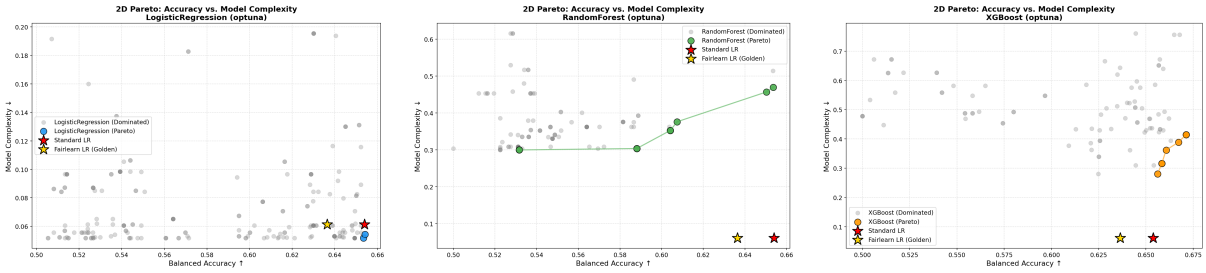
(a) Logistic Regression (b) Random Forest (c) XGBoost

Figure 3: Pymoo NSGA-II: Balanced Accuracy vs. Model Complexity across three model types.



(a) Logistic Regression (b) Random Forest (c) XGBoost

Figure 4: Optuna NSGA-II: Balanced Accuracy vs. Demographic Parity Violation across three model types.



(a) Logistic Regression (b) Random Forest (c) XGBoost

Figure 5: Optuna NSGA-II: Balanced Accuracy vs. Model Complexity across three model types.

4.3 3D Pareto Frontier

The complete 3D Pareto surface provides the most comprehensive view of the objective space. (Note: The pipeline generates an interactive HTML Plotly graph; a representative screenshot is provided below).

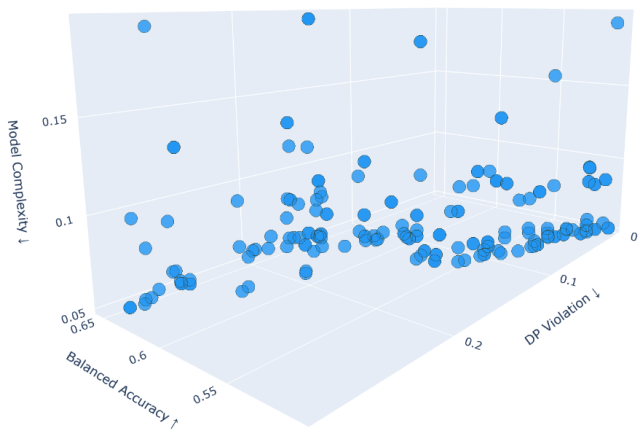


Figure 6: 3D Scatter Plot of the Pareto Frontier (Balanced Accuracy vs DP Violation vs Model Complexity in Logistic Regression).

5 Conclusion

This study validates the necessity of multi-objective optimization in algorithmic fairness. By evaluating over 20,000 hyperparameter combinations across two state-of-the-art frameworks (Pymoo and Optuna), we successfully mapped the Pareto frontiers for Logistic Regression, Random Forest, and XGBoost models.

Our findings indicate that achieving near-zero Demographic Parity violations inevitably incurs a cost in predictive accuracy. Furthermore, highly complex models do not linearly guarantee better fairness-accuracy trade-offs; often, simpler models (lower depth, higher regularization) provide identical fairness constraints with significantly lower complexity overheads. The framework developed in this project provides a scalable, automated pipeline for policymakers and data scientists to explicitly select models based

on defined ethical and performance tolerances.